14 Implement House price Prediction using appropriate machine learning algorithm

Code:

```
import matplotlib.pyplot as plt
from sklearn.neighbors import KNeighborsRegressor

# House area in square feet
X = [[500], [700], [900], [1100], [1300], [1500]]

# House prices in lakhs
y = [10, 14, 18, 22, 26, 30]

# Train KNN Regression model
model = KNeighborsRegressor(n_neighbors=3)
model.fit(X, y)

# Predict price for a 1000 sq. ft. house
area = [[1000]]
prediction = model.predict(area)

print("KNN Regression")
print("House Area:", area[0][0], "sq. ft.")
print("Predicted House Price:", round(prediction[0], 2), "lakhs")

# Plot graph
plt.scatter(X, y, label="Actual Prices")
plt.scatter(area, prediction, marker="X", s=200,
            label="Predicted Price")

plt.xlabel("House Area (sq. ft.)")
plt.ylabel("House Price (Lakhs)")
plt.title("KNN Regression - House Price Prediction")
plt.legend()
plt.grid(True)
plt.show()
```

Output:

```
KNN Regression
House Area: 1000 sq. ft.
Predicted House Price: 18.0 lakhs
```

